WORKED SOLUTION

STEP III 1998, Question 13

Probability and Statistics • Maximum Likelihood Estimation

STEP III 1998, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q13

The key idea

Here the unknown is n, and it is an integer, so calculus is no use: we can't differentiate with respect to n. The standard tool for maximising a sequence is to look at the *ratio* of consecutive terms. The sequence rises while the ratio exceeds 1 and falls once it drops below 1, and a tie (two MLEs) happens exactly when the ratio equals 1 at some point.

Solution

The number of heads in n tosses of a fair coin is binomial, so

$$\mathrm{P}(k \text{ heads in } n \text{ tosses}) = \binom{n}{k}\frac{1}{2^n}, \qquad n \geq k,$$

and the probability is 0 if $n < k$. Throughout we take $k \geq 1$ (the friend has seen at least one head), so only $n \geq k$ is a candidate.

The general fair-coin calculation

Write $L(n) = \binom{n}{k}2^{-n}$ for $n \geq k$, thought of as a function of n with k fixed. For $n \geq k$,

$$\frac{L(n+1)}{L(n)} = \frac{\binom{n+1}{k}}{\binom{n}{k}} \cdot \frac{1}{2} = \frac{n+1}{n+1-k} \cdot \frac{1}{2},$$

using $\binom{n+1}{k} = \frac{n+1}{n+1-k}\binom{n}{k}$. Since $n+1-k > 0$, we can clear the denominator without reversing anything:

$$\frac{L(n+1)}{L(n)} \geq 1 \iff n+1 \geq 2(n+1-k) \iff n \leq 2k-1,$$

with equality exactly when $n = 2k-1$. So

$$L(k) < L(k+1) < \cdots < L(2k-1) = L(2k) > L(2k+1) > L(2k+2) > \cdots.$$

Ask yourself How does the ratio tell us where the maximum is?

For $n < 2k-1$ the ratio is greater than 1, so each term is bigger than the one before: the sequence strictly increases up to $n = 2k-1$. At $n = 2k-1$ the ratio is exactly 1, so $L(2k) = L(2k-1)$. For $n > 2k-1$ the ratio is below 1 and the sequence strictly decreases from then on. Hence the largest value is attained at exactly two values of n.

One head

With $k = 1$, the two MLEs are $n = 2k-1 = 1$ and $n = 2k = 2$. Directly,

$$L(1) = \tfrac{1}{2}, \quad L(2) = 2 \cdot \tfrac{1}{4} = \tfrac{1}{2}, \quad L(3) = 3 \cdot \tfrac{1}{8} = \tfrac{3}{8},$$

and the argument above shows $L(n)$ keeps decreasing after $n = 2$. So there are two MLEs, 1 and 2, as required.

k heads

By the general calculation, the two MLEs of the number of tosses are

$$n = 2k-1 \quad \text{and} \quad n = 2k.$$

Biased coin

Now let $0 < p < 1$ and $q = 1-p$, so that

$$L(n) = \binom{n}{k} p^k q^{n-k}, \qquad n \geq k.$$

For $n \geq k$, the factor p^k cancels and one extra factor q appears:

$$\frac{L(n+1)}{L(n)} = \frac{(n+1)q}{n+1-k}.$$

Again $n+1-k > 0$, so

$$\frac{L(n+1)}{L(n)} \gtreqless 1 \iff (n+1)(1-p) \gtreqless n+1-k \iff k \gtreqless (n+1)p \iff n+1 \lesseqgtr \frac{k}{p}.$$

So L increases while $n+1 < k/p$, is level when $n+1 = k/p$, and decreases once $n+1 > k/p$. Note that $k/p > k$, since $p < 1$, so the turning point lies inside the range $n \geq k$.

- If k/p is *not* an integer, let $m = \lfloor k/p \rfloor$. Then $n+1 < k/p$ exactly when $n+1 \leq m$, so $L(k) < \cdots < L(m)$, and $L(m+1) < L(m)$ with the decrease continuing. There is a single MLE,
$$n = \left\lfloor \frac{k}{p} \right\rfloor.$$

- If $k/p = N$ is an integer, then L increases up to $L(N-1)$, has $L(N) = L(N-1)$, and then decreases. There are two MLEs,

$$n = \frac{k}{p} - 1 \quad \text{and} \quad n = \frac{k}{p}.$$

  Both are at least k: since N is an integer with $N > k$, we have $N - 1 \geq k$.

So the MLE is unique **if and only if k/p is not an integer**. For a fair coin $k/p = 2k$ is always an integer, which is why the earlier parts always gave two MLEs, $2k-1$ and $2k$.

Checks

- The answer makes sense: with k heads and head probability p, we'd expect about k/p tosses, and the MLE is k/p rounded down (or the tie either side when it is an integer).
- Take $p = \frac{1}{3}$, $k = 1$, so $k/p = 3$. Then $L(2) = 2 \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{4}{9}$ and $L(3) = 3 \cdot \frac{1}{3} \cdot \frac{4}{9} = \frac{4}{9}$, a tie, while $L(4) = 4 \cdot \frac{1}{3} \cdot \frac{8}{27} = \frac{32}{81} < \frac{4}{9}$.
- A computer computed $L(n)$ exactly with fractions and found the maximisers for $k = 1, \ldots, 39$ with a fair coin (always $\{2k-1, 2k\}$) and for 924 pairs (p, k) with p a fraction of denominator up to 12 and $k \leq 14$. Every case agreed with the rule above.

Common slips

- Differentiating with respect to n. Here n is an integer and $\binom{n}{k}$ isn't defined for other values of n without extra machinery; the ratio test is the right tool.
- Getting the inequality the wrong way round, or forgetting that equality of the ratio with 1 is what produces the second MLE.
- Answering $\lfloor k/p \rfloor$ in every case and missing the tie when k/p is an integer.

Beyond STEP

Maximum likelihood is the central method of estimation in statistics: choose the parameter value that makes the observed data most probable. Usually the parameter is continuous and we maximise the log-likelihood by calculus. Estimating an integer like n (the "number of trials" problem) is a classic case where it fails, and this question shows that the MLE need not even be unique. Blitzstein and Hwang, *Introduction to Probability*, is good on the binomial distribution, and Grimmett and Welsh, *Probability: An Introduction*, covers the background probability.

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